CHENGHAO XU

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EDUCATION

Southern University of Science and Technology Shenzhen, China
- B.Eng in Mechanical Engineering (**with Honors**) *Sep.2016 - Jun.2020*
  Overall GPA: **3.65** / 4.00, Major GPA: **3.85** / 4.00
- ***Courses:*** *CAD Engineering Drawing, Machine Design, Theoretical Mechanics, Advanced Manufacturing*

Delft University of Technology Delft, Netherlands
- MSc Robotics (GPA: 8.3/10) *Sep.2020 - Nov.2023*
- ***Courses:*** *Dynamics & Control, Planning & Decision Making, Machine Perception, Deep Learning*

The University of British Columbia Vancouver, Canada
- Vancouver Summer Program (Grades: A-) *Jul.2017 - Aug.2017*

PROFESSIONAL EXPERIENCE

GRADE: Generating Realistic Animated Dynamic Environment
Student Research Assistant *Advisor: Prof. Aamir Ahmad*
Max Planck Institute for Intelligent Systems, Germany *Aug.2022 – Mar.2022*
- Developed synthetic environment's generation framework for robot navigation in **dynamic scenes** based on NVIDIA Issac Sim, involving **optical flow ground truth, motion blur, sensor noise**, and **event streams**
- Conducted verification of framework's generalization and usability with comparative testing on state-of-the-art SLAM frameworks (RTAB-Map, ORB-SLAM2, DynaSLAM, Dynamic-VINS, TartanVO, StaticFusion)
- Generated synthetic blurry image datasets with ground truth labels to fine-tune YOLO and Mask R-CNN models on human detection tasks, resulting in improvements of ~5% on YOLO and 2% on Mask R-CNN

Multi-Camera Real-Time Surveillance Video Stitching
Computer Vision Engineer *Internship*
Lely Technologies, Netherlands *Feb.2022 - Jul.2022*
- Investigated feature-based image stitching methods for multi-view cameras with parallax, and conducted a comparative analysis of **ORB, BRISKS, SIFT and VGG16 features** for image stitching
- Proposed a feature matching strategy based on AutoStitch framework by mapping original features into undistorted images within corresponding ROIs separately to improve overall stitching performance
- Deployed the real-time stitching systems and positioning system in ROS2 in the real world

Control Engineer for Industrial Manipulator *Internship*
ROKAE Robotics, China *Jul.2021 - Sep.2021*
- Developed online trajectory planning algorithm for industrial robots based on the **discrete-time double S profile**, which achieves trajectory update in real-time with **continuous acceleration**
- Accomplished trajectory planning and optimization under multiple industrial scenarios, and simulated joint states (peak torque, angular speed, power) during execution using Robotics System Toolbox
- Investigated impact of DH parameter deviation on the end effector position accuracy by Robotics Toolbox

Quadruped Robot based on Discontinuous Terrain Perception
Student Research Assistant *Advisor: Dr. Jianwen Luo*
SUSTech Institute of Robotics, China *Jun.2019 - Dec.2019*
- Accomplished and optimized the design of robot structure by SolidWorks to obtain higher stability
- Constructed prototype and adjusted motor control to achieve basic stepping or walking locomotion
- Secured funding from Youth Program of National Natural Science Foundation of China

Upper-Limb Passive Adaptive Assisted Exoskeleton

Research & Development Intern *Advisor: Prof. Yiming Rong*

Milebot Robotics, China *Jun.2019 - Sep.2019*

- Conducted market analysis on exoskeleton products and used decision matrix to generate design concepts
- Optimized structural design of the energy storage and adaptive device to achieve adjustable assist effect
- Accomplished assist effect through mechanical analysis and numerical simulations using MATLAB
- Investigated the secondary effect of EMG signal and oxygen consumption in working environment

PROJECT EXPERIENCE

TIAGo Robot for Expiring Items Picking in Retail Environment

Best Project Award *with Ahold Delhaize* *Apr.2022 - Jun.2022*

- Constructed **ROS behavior tree** architecture containing motion planning module, which dynamically adjusts intermediary goals and performs items picking/placing in sequence.
- Developed the indoor localization method based on Apriltag and indoor exploration algorithm
- Deployed the real-time YOLO v5 model to detect items in store, which reaches a mAP of 0.7

Autonomous Navigation for Site Inspection with Spot Robot

Part-time at Yes!Delft, Netherlands *Mar.2022 - Jun.2022*

- Deployed autonomous inspection module based on Spot SDK including Mapping and Waypoint Planning
- Designed the functional GUI based on PyQt5 and Open3D to accomplish **Camera View Navigation and Visualization** remotely from the point cloud captured by Spot

Path Planner for Quadrotor based on RRT* and k-PRM Methods

Course Project *Nov.2021 - Jan.2022*

- Deployed RRT* and k-PRM path planner to generate collision-free path on 3D random obstacle map
- Accomplished trajectory optimization using minisnap and corridor bounding method

Manufacture of Aluminum Hollow Nested Cubes

Course Project *Feb.2019 - May.2019*

- Accomplished the structural design and machining simulation of aluminum nested cubes
- Delivered process planning with the skilled operation of CNC machining center and wire-cut EDM

PUBLICATIONS & PRESENTATIONS

- **C. Xu**, E. Bonetto, A. Ahmad, "DynaPix-SLAM: A Pixel-based Dynamic SLAM Approach", *arXiv preprint*, 2023.
- E. Bonetto, **C. Xu**, A. Ahmad, "GRADE-RR: Generate Realistic Animated Dynamic Environments for Robotics Research," Submitted to *IEEE Transactions on Robotics (T-RO)*, 2023. [Link]
- H. Liu, K. Fang, L. Chen, **C. Xu**, et al., "Implementation of a Long-lasting, Untethered, Lightweight, Upper-limb Exoskeleton", Submitted to *IEEE/ASME Transactions on Mechatronics (T-MECH)*, 2023
- E. Bonetto, **C. Xu**, A. Ahmad, "Learning from Synthetic Data Generated with GRADE," *In Proceedings of ICRA 2023 Workshop on Pretraining for Robotics*, 2023. [Link]
- E. Bonetto, **C. Xu**, A. Ahmad, "Simulation of Dynamic Environments for SLAM," *In Proceedings of ICRA 2023 Workshop on Active Methods in Autonomous Navigation,* 2023. [Link]
- L. Chen, **C. Xu**, H. Liu, etc., "Design and Verification of Passive Upper Arm Assisted Exoskeleton," *Presented at Annual Conference of Chinese Robotics Society (CCRS)*, 2019.
- **C. Xu**, "Breaking the Wall of Intensive Work Above Head," *Oral presented at Falling Walls Lab Berlin*, 2019.
- L. Chen, H. Liu, **C. Xu**, etc., Assisted exoskeleton devices. *(Patent No. ZL201921545381.6)*

AWARDS & HONORS

- 2016 Freshmen Scholarship | *Excellence Award*
- 2016 The 2nd Shenzhen College Students Thinking and Debate Contest | *Champion*
- 2017 The 3rd Shenzhen College Students Thinking and Debate Contest | *Third Place,* ***Best Debater***
- 2018 National Scholarship Award | *Nomination Prize (**Top 1%**)*
- 2018 Merit Student Scholarship for Exceptional Performance | *First Class (**Top 5%**)*
- 2018 Dean's Award for Academic Excellence in College *(**Top 1%**)*
- 2018 National College Student Science Contest on Energy Saving & Emission Reduction | *Third Prize*
- 2019 "TI Cup" National Undergraduate Electronics Design Contest | *Provincial Second Prize*
- 2019 Guangdong Entrepreneurship and Innovation Competition | *Excellence Award*
- 2019 Falling Walls Lab Shenzhen | *First Prize*
- 2019 Merit Student Scholarship for Exceptional Performance | *First Class (**Top 5%**)*
- 2019 National Scholarship Award | *Nomination Prize (**Top 1%**)*
- 2020 "Challenge Cup" Provincial College Student Entrepreneurship Plan Competition | *Gold Award*
- 2020 "Challenge Cup" National College Student Entrepreneurship Plan Competition | *Bronze Award*
- 2020 Excellent Graduate in Southern University of Science and Technology *(**Top 5%**)*
- 2022 Erasmus+ Traineeship Grants

EXTRACURRICULAR ACTIVITIES

Secretary of SUSTech Student Union *Sep.2016 - Jun.2017*

- Organized school cultural activities and coordinated the mission of various departments

Senior Counselor for Freshmen of SUSTech *Jun.2017 - Jun.2019*

- Advised 60 freshman students via a series of mini-lectures and activities
- Established collaborative and caring community culture, solved daily problems and arranged social activities

Debate Team Leader of SUSTech *Feb.2018 - Feb.2019*

- Delivered daily training for members in competition and maintained communication with other organizations

Volunteer of The Asia-Pacific Regional Seminar on MOOCs for Higher Education *Jun.2018*

Volunteer of China Hardware Innovation Camp (CHIC2019) *Jul.2019*

- Launched by École Polytechnique Fédérale de Lausanne (EPFL)
- Participated in the project ***Heptabox*** to improve the effectiveness of medication intake

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, C/C++, HTML/CSS
- **Software:** ROS, OpenCV, Open3D, Pytorch, TensorFlow, g2o, Git, LaTeX, SolidWorks
- **Hardware:** Arduino, Mechanical Design, Machining (Milling, Turning, Wire-Cut EDM), 3D Printing